

VERMILION REGIONAL, IL, USA

WMO: 722076

Lat: 40.200N

Lon: 87.600W

Elev: 664

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	5.0	-7.8	3.7	1.4	-2.9	4.8	6.1	28.2	33.1	26.0	34.2	6.5	300	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	91.9	73.9	89.6	74.2	86.8	73.3	79.2	87.5	77.4	85.3	75.8	83.2	9.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	143.7	84.5	75.0	134.6	82.0	73.3	126.7	80.3	43.3	87.6	41.2	85.6	39.7	83.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.7	19.0	DB	-8.7	94.3	5.8	3.3	-12.9	96.7	-16.3	98.6	-19.6	100.4	-23.8	102.8	
			WB	-8.4	82.5	6.4	1.3	-13.0	83.4	-16.8	84.2	-20.4	84.9	-25.0	85.8	

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	26.6	29.0	41.2	53.2	64.5	72.4	74.2	72.2	67.4	55.1	42.3	32.1	(6)
(7)		DBStd	19.50	12.59	12.06	12.03	9.63	9.01	5.83	5.43	5.55	7.44	9.62	10.03	10.47	(7)
(8)		HDD50	2619	729	593	331	76	6	0	0	0	54	270	560		(8)
(9)		HDD65	5555	1191	1008	744	369	125	10	4	8	60	334	681	1021	(9)
(10)		CDD50	3580	3	5	58	171	456	673	750	687	521	213	39	4	(10)
(11)		CDD65	1042	0	0	6	14	111	233	289	230	130	28	1	0	(11)
(12)		CDH74	9670	0	0	45	174	1017	2042	2749	2100	1277	265	1	0	(12)
(13)		CDH80	3257	0	0	6	25	318	681	990	702	463	72	0	0	(13)
(14)	Wind	WSAvg	8.2	10.5	9.9	10.0	10.5	8.7	7.1	5.1	4.5	5.4	7.7	9.4	9.4	(14)
(15)	Precipitation	PrecAvg	38.9	2.1	1.9	3.0	3.9	4.2	4.2	4.4	3.7	2.9	3.0	3.2	2.6	(15)
(16)		PrecMax	54.3	7.3	6.3	6.4	10.9	10.0	12.4	12.8	9.0	8.7	8.6	10.2	7.0	(16)
(17)		PrecMin	24.0	0.1	0.4	0.8	0.6	0.2	0.3	0.6	1.2	0.0	0.3	0.5	0.2	(17)
(18)		PrecStd	6.5	1.5	1.2	1.5	2.2	2.1	2.4	2.7	1.9	1.9	2.0	1.9	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	63.9	80.3	82.4	90.7	92.4	96.7	94.0	93.5	87.8	72.1	62.1	(19)
(20)		MCWB	58.6	55.8	65.1	60.7	65.0	73.3	76.3	77.3	71.3	72.2	57.7	58.1		(20)
(21)		2%	DB	55.0	57.4	72.7	79.2	87.0	90.0	90.9	90.3	89.9	81.1	66.5	55.4	(21)
(22)		MCWB	51.6	53.5	61.7	60.8	68.4	73.2	76.8	75.0	73.6	65.4	58.0	52.1		(22)
(23)		5%	DB	49.8	52.0	65.6	74.8	83.7	87.3	88.3	86.6	85.6	75.6	61.7	51.6	(23)
(24)		MCWB	46.4	47.8	57.3	59.8	66.6	73.1	76.4	74.3	70.7	63.3	55.6	49.0		(24)
(25)		10%	DB	43.4	45.8	60.3	70.4	79.9	83.9	85.6	83.9	81.4	71.5	57.1	46.1	(25)
(26)		MCWB	40.1	42.0	53.0	58.3	65.5	70.7	74.8	73.3	67.6	62.3	51.0	43.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	58.8	66.0	67.3	74.1	79.3	82.0	80.8	77.3	73.9	64.0	59.1	(27)
(28)		MCDWB	62.1	61.8	77.0	77.0	83.6	87.4	88.9	90.7	87.9	84.5	68.3	61.5		(28)
(29)		2%	WB	52.4	53.2	62.6	64.4	71.9	77.0	80.0	78.2	75.1	70.1	59.7	53.1	(29)
(30)		MCDWB	54.6	56.6	71.5	73.3	81.0	85.6	87.9	86.7	84.9	77.0	64.4	55.0		(30)
(31)		5%	WB	46.7	47.9	59.0	62.1	70.2	75.2	78.0	76.5	73.1	66.9	56.0	48.4	(31)
(32)		MCDWB	48.9	51.0	64.8	70.8	78.7	83.1	85.5	84.0	81.2	72.3	61.3	50.4		(32)
(33)		10%	WB	40.7	42.1	54.2	59.7	68.4	73.4	76.3	74.8	71.1	63.0	51.9	43.7	(33)
(34)		MCDWB	43.7	46.0	59.2	68.8	76.6	80.8	83.1	81.5	78.1	69.7	56.5	46.1		(34)
(35)	Mean Daily Temperature Range	MDBR	14.5	15.6	17.9	21.0	20.3	19.5	19.7	21.1	23.1	20.9	16.6	13.3		(35)
(36)		5% DB	MCDBR	20.0	22.1	23.1	26.5	24.9	21.9	21.9	23.6	27.0	26.5	22.3	19.0	(36)
(37)		MCWBR	18.0	18.4	14.4	13.7	10.8	10.2	10.8	12.2	11.7	13.4	14.9	16.9		(37)
(38)		5% WB	MCDWB	19.3	21.4	21.2	22.0	20.3	18.4	18.9	19.6	21.8	21.5	20.0	18.2	(38)
(39)		MCWBR	18.5	19.1	14.2	14.3	10.8	10.2	11.1	11.4	10.8	12.8	15.6	17.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.317	0.353	0.398	0.435	0.439	0.456	0.458	0.421	0.365	0.319	0.299		(40)
(41)		taud	2.433	2.377	2.360	2.241	2.192	2.220	2.162	2.149	2.230	2.381	2.468	2.456		(41)
(42)		Ebn at Noon	274	285	285	278	269	267	261	257	259	263	265	265		(42)
(43)		Edh at Noon	25	31	35	43	46	44	47	46	40	31	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	560	841	1158	1500	1712	1966	1939	1735	1453	997	663	468		(44)
(45)		RadStd	60	84	89	112	148	117	104	101	106	99	65	55		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page